

for waste management. Their market is expected to witness a CAGR of 21.5 per cent by 2024. Reasons for such an alarming growth are:

- Robotics continuously reducing the cost of trash-bots
- Increased use of the Internet of Thing (IoT) technology, consumer electronics and connected devices
- Need for efficient waste-management systems
- Concept of smart cities

Trash-bots can easily sort the garbage, and increase productivity and efficiency. Visual sensors, metal detectors, weight-measurement sensors, spectrometers, 3D laser scanners and touch sensors are important components of trash-bots. These components can collect huge amounts of valuable data needed for recycling industries and act accordingly.

Connecting e-waste to waste management

Less than five per cent of India's total electronic waste, or e-waste, gets recycled due to the absence of proper infrastructure, legislation and framework for disposing of electronic gadgets and products that have reached their end of life. According to Associated Chambers of Commerce and Industry of India (ASSOCHAM), India generates over 440,000 tonnes of e-waste annually (based on a 2012 survey). This is growing at a rate of about 20 per cent annually. About half of all unused and end-of-life electronic products lies in landfills, junkyards and warehouses.

Over 90 per cent of the e-waste generated in India is managed by the unorganised sector, and scrap dealers in this market dismantle the disposed-of products instead of recycling them. A number of startups have entered into e-waste management since the concept of smart cities and Make in India started. Some startups engaged in e-waste management are mentioned below.

- Altero Electronics Asset Management Co., which is mainly concerned with e-waste mining.
- Shivalic Solid Waste Management is an offshoot of UPL Group of Companies, Mumbai. It provides several waste-management services, including environmental monitoring, and e-waste, CFL and used lead-acid batteries recycling, among others.
- Namo eWaste Management Ltd was started by Akshay Jain, a 28-year-old entrepreneur from New Delhi. The company processes up to 20 tonnes of e-waste daily. With collection centres across 12 states and union territories across the country, the startup is building strategic partnerships with leading electronic companies.

The road ahead

In recent years, a large number of startups have ventured into the field of waste management. This number is expected to grow further. Such technologies as AI, Big Data, machine learning and robotics, if implemented properly, can make the Indian waste management sector much more efficient. **EFY**



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